Fire Detecting System – General Data

1. General

A fire detection system is provided for each engine. In the event of an engine fire, thermal detector units installed in the engine nacelle actuate and complete an electrical circuit to illuminate a red warning light for the appropriate fire extinguisher push button, located on the top of the instrument panel in the flight compartment. Annunciation is available on Apex as well as the Primary Flight Displays (PFDs), Crew Alerting System (CAS) and an aural warning.

2. Description

Each engine fire detecting system comprises four thermal units connected to two wire loops, a fire warning light, magnetic and thermal circuit breakers, the associated electrical wiring, and a circuit test switch which is common to both engine systems. Aural warning consisting of a tone, the word “FIRE” is annunciated, and Left and/or Right Engine Fire warning CAS messages are shown on the systems MFD. The location of all system components is shown on Figure 1.

A. Fire Detecting Thermal Units

The four fire detecting thermal units for each engine, and their connecting wire loops, are located on the forward and rear fireseals in the engine nacelle, and on a nacelle frame at station 54.60. The two forward thermal units actuate at a temperature of 450°F, and the two rear thermal units at a temperature of 300°F.

B. Fire Detection Warning Lights and Switches

The fire detection lights are integral with the fire extinguisher push button located on the top of the instrument panel, the warning light/push button are annotated LEFT ENGINE and RIGHT ENGINE. Four circuit breakers, two for each engine system, are provided on the circuit breaker panel; two are thermal circuit breakers labeled FIRE DETECTION L and R, and two are magnetic circuit breakers labeled FIRE DET FAULT IND L and R.

C. Fire Detecting System Magnetic Circuit Breakers

Two of the system circuit breakers, one in each engine circuit, are of the magnetic type, having a single set of switch contacts with a center moving arm mechanically linked to its magnetic coil.

3. Operation

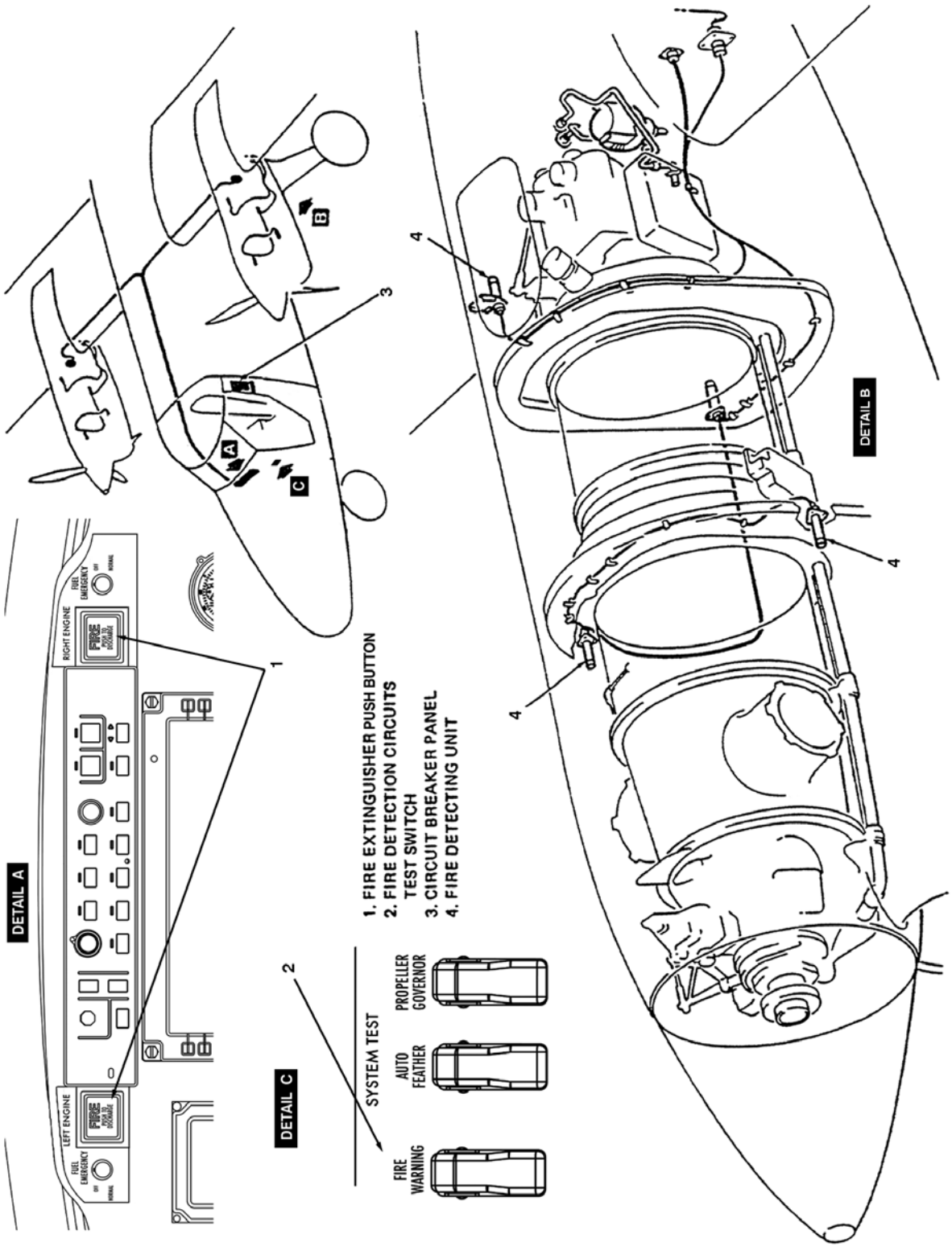
With the aircraft 28 Volt DC bus bars energized and the system circuit breakers engaged, the magnetic circuit breaker in each engine circuit supplies power to one of the two thermal detecting loops. Simultaneously the center arm of the magnetic circuit breaker, which is connected to the second of the two thermal detecting loops through the warning lamp, moves to the ground position.

When the contacts within anyone of the thermal units is actuated (due to a fire condition) the contacts close and complete a circuit between the two loops, so causing the warning lamp to illuminate and the “FIRE” warning is annunciated.

In the event of a ground fault (short circuit) occurring in the loop supplied through the magnetic circuit breaker, the circuit breaker will disengage, the center arm within the breaker moves to disconnect the ground and provide an alternative power supply to the second loop.

The TEST switch when selected to TEST, connects the two thermal-detecting loops of each engine system in series and completes a circuit through the warning light to ground, thus allowing the continuity of the loops, and the serviceability of the lights to be verified.

The aural warning is continuous except when muted by depressing the MASTER WARNING switch.



Fire Detection System – Component Locations
Figure 1

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4. Adjustment/Test

A. Operational Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Engage FIRE DETECTION L and R and FIRE DET FAULT IND L and R circuit breakers.

3) Select FIRE WARNING switch to up position and check warning lights in both FIRE warning lights/push buttons illuminate.

4) Short out one of the overheat sensors in the LH nacelle by placing a suitable jumper across the two terminals of the sensor and check that the left engine fire warning light/push button is illuminated. Remove jumper and replace rubber cap over sensor terminals.

5) Repeat Para 4.A.4) at the RH nacelle and check that the right engine FIRE warning light/push button illuminates.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

7) Disengage FIRE DETECTION L and R circuit breakers.

B. Bench Test Fire Detecting Thermal Units

Equipment required:

- Fenwall field test kit 80001–0 or equivalent
- General Radio 500-volt AC Megger or equivalent

1) Test Insulation Resistance

a) Connect a General Radio 500-volt Megger or equivalent between the shell of fire detecting unit and both its leads tied together.

b) Check that minimum insulation resistance is 3 Megohms.

2) Test Control Set Point

a) Install the fire detecting unit in the test block of Fenwall field test kit 80001–0 or equivalent in a manner simulating as closely as possible the normal installation of the detecting units (especially in regard to restraint by clamps, brackets and external connections).

b) Set ON–OFF switch to OFF and connect test unit to 115-volt AC outlet.

c) Set INVERSE–REGULAR switch to REGULAR.

d) Set HIGH–LOW switch to HIGH.

e) Place thermometer in appropriate well in test block.

f) Connect receptacle leads to the fire detecting unit leads.

g) Set ON–OFF switch to ON and wait 15 minutes after cycling begins for system to reach a stable condition.

h) Calculate the control set point by taking the average of three consecutive readings of test block temperature at the instant of contact closure during stable operation.

Note All temperature readings are to be made at the instant the heater pilot lamps come “on”.

i) Check that tolerances are within ± 5°F or 2% of the designated control set point, whichever is the greater.

Note PDM17343–62–300 Control set point is 300°F.
PDM17343–62–450 Control set point is 450°F.

j) If unit is not within tolerances above, replace with serviceable unit.